

Crystal Structure Comparison: FCC, BCC, HCP, Simple Cubic, and

Quick Comparison Table

SC

Full name: Simple cubic
Conventional unit cell contents: 1 atom/cell
Coordination number: 6
Atomic packing factor: 0.52
Close-packed plane: None truly close-packed; densest plane is {100}
Close-packed direction: $\langle 100 \rangle$ directions are nearest-neighbor directions
Stacking / key geometry: Simple cubic layers stacked directly above each other
Common examples: Polonium, idealized model

BCC

Full name: Body-centered cubic
Conventional unit cell contents: 2 atoms/cell
Coordination number: 8 nearest neighbors
Atomic packing factor: 0.68
Close-packed plane: None truly close-packed; densest planes are {110}
Close-packed direction: $\langle 111 \rangle$ are closest-packed directions
Stacking / key geometry: Atoms touch along body diagonal
Common examples: alpha-Fe, Cr, W, Mo

FCC

Full name: Face-centered cubic
Conventional unit cell contents: 4 atoms/cell
Coordination number: 12
Atomic packing factor: 0.74
Close-packed plane: {111}
Close-packed direction: $\langle 110 \rangle$
Stacking / key geometry: ABCABC close-packed stacking
Common examples: Al, Cu, Ni, Ag, Au, Pb

HCP

Full name: Hexagonal close-packed
Conventional unit cell contents: 6 atoms/conventional cell; 2 atoms/primitive cell
Coordination number: 12
Atomic packing factor: 0.74 ideal
Close-packed plane: {0001} basal plane
Close-packed direction: $\langle 11\bar{2}0 \rangle$
Stacking / key geometry: ABAB close-packed stacking
Common examples: Mg, Ti, Zn, Co

Rock Salt

Full name: NaCl-type ionic structure
Conventional unit cell contents: 4 formula units/cell; 4 cations + 4 anions
Coordination number: 6 for each ion, opposite charge only
Atomic packing factor: Depends on ion sizes; not a single metallic APF
Close-packed plane: Not usually described by metallic close-packed planes; each ion sublattice is FCC with {111} close-packed layers
Close-packed direction: Nearest unlike-ion directions are $\langle 100 \rangle$
Stacking / key geometry: Two interpenetrating FCC sublattices; cations fill all octahedral sites of anion FCC lattice
Common examples: NaCl, MgO, CaO, FeO

Main Takeaways

FCC and HCP are the two classic close-packed metallic structures. They both have:

- coordination number 12
- maximum equal-sphere packing efficiency of 0.74
- triangular/hexagonal close-packed atomic layers

The difference is the stacking sequence:

FCC: ABCABC...
HCP: ABAB...

BCC is dense, but it is not close-packed. It has close-packed directions, especially $\langle 111 \rangle$, but no plane has the true triangular close-packed arrangement found in FCC {111} or HCP {0001}.

Simple cubic is much more open. Its atoms touch along cube edges, but it does not contain a true close-packed plane.

Rock salt is an ionic crystal, so it should not be treated exactly like a one-atom metallic lattice. It is best understood as two interpenetrating FCC lattices: one for anions and one for cations.

FCC: Face-Centered Cubic

In FCC, atoms are located at the 8 corners and the 6 face centers of the cubic unit cell.

Effective atoms per conventional unit cell:

$$8 \text{ corners} \times 1/8 + 6 \text{ face centers} \times 1/2 = 4 \text{ atoms}$$

Key facts:

- Coordination number: 12
- Atomic packing factor: 0.74
- Close-packed planes: {111}
- Close-packed directions: $\langle 110 \rangle$
- Stacking sequence: ABCABC...
- Atoms touch along the face diagonal.

Radius-lattice parameter relation:

$$4r = \sqrt{2}a$$
$$a = 2\sqrt{2}r$$

The FCC {111} plane has a triangular arrangement of atoms. This is why it is a close-packed plane.

Common FCC metals:

- Al
- Cu
- Ni
- Ag
- Au
- Pb

HCP: Hexagonal Close-Packed

HCP is also a close-packed structure, but it uses a hexagonal unit cell instead of a cubic one.

Key facts:

- Coordination number: 12
- Atomic packing factor: 0.74 for ideal HCP
- Close-packed plane: {0001}, also called the basal plane
- Close-packed directions: $\langle 11\bar{2}0 \rangle$
- Stacking sequence: ABAB...

Ideal HCP geometry:

$$a = 2r$$
$$c/a = 1.633$$

HCP and FCC have the same packing efficiency and coordination number. The major difference is the layer stacking:

HCP repeats every 2 layers: ABAB...
FCC repeats every 3 layers: ABCABC...

Common HCP metals:

- Mg
- Ti
- Zn
- Co

BCC: Body-Centered Cubic

In BCC, atoms are located at the 8 corners and 1 body center of the cubic unit cell.

Effective atoms per conventional unit cell:

$$8 \text{ corners} \times \frac{1}{8} + 1 \text{ body center} = 2 \text{ atoms}$$

Key facts:

- Coordination number: 8 nearest neighbors
- Atomic packing factor: 0.68
- No true close-packed plane
- Densest planes: {110}
- Closest-packed directions: $\langle 111 \rangle$
- Atoms touch along the body diagonal.

Radius-lattice parameter relation:

$$4r = \sqrt{3}a$$

$$a = 4r / \sqrt{3}$$

Important exam note:

Although BCC often slips in $\langle 111 \rangle$ directions and has dense {110} planes, the {110} planes are not close-packed in the same strict sense as FCC {111} or HCP {0001}. A true close-packed plane must have atoms arranged in a triangular/hexagonal net.

Common BCC metals:

- alpha-Fe
- Cr
- W
- Mo

Simple Cubic

In simple cubic, atoms are located only at the 8 corners of the cubic unit cell.

Effective atoms per conventional unit cell:

$$8 \text{ corners} \times 1/8 = 1 \text{ atom}$$

Key facts:

- Coordination number: 6
- Atomic packing factor: 0.52
- No true close-packed plane
- Densest plane: {100}
- Nearest-neighbor directions: $\langle 100 \rangle$
- Atoms touch along cube edges.

Radius-lattice parameter relation:

$$a = 2r$$

Simple cubic is relatively open compared with FCC, HCP, and BCC. It is uncommon among real elemental solids.

Rock Salt Structure

Rock salt, also called the NaCl structure, is an ionic crystal structure. It is not a simple metallic packing of identical atoms.

The easiest way to describe it:

One ion type forms an FCC lattice.
The other ion type occupies all octahedral sites.

For NaCl specifically:

- Cl⁻ ions can be viewed as forming an FCC sublattice.
- Na⁺ ions occupy all octahedral holes.
- Equivalently, Na⁺ and Cl⁻ form two interpenetrating FCC sublattices.

Key facts:

- Coordination number: 6 for Na⁺ and 6 for Cl⁻
- Geometry around each ion: octahedral
- Formula units per conventional unit cell: 4 NaCl
- Ion positions are commonly described as FCC anions plus cations in octahedral sites.
- Nearest unlike-ion directions: $\langle 100 \rangle$

Nearest-neighbor relation:

$$\begin{aligned} r_{\text{cation}} + r_{\text{anion}} &= a/2 \\ a &= 2(r_{\text{cation}} + r_{\text{anion}}) \end{aligned}$$

Radius ratio condition for stable octahedral coordination:

$$0.414 \leq r_{\text{cation}} / r_{\text{anion}} \leq 0.732$$

Close-packed plane note:

Rock salt is often related to FCC packing because the larger ions, commonly the anions, form an FCC array. Therefore the anion sublattice has FCC-style close-packed {111} planes. However, for the full NaCl ionic structure, it is usually better to describe the structure by coordination number and octahedral coordination rather than saying it has a metallic close-packed plane.

Exam-Friendly Distinctions

Do not confuse these ideas:

Close-packed plane | A plane where atoms form a triangular/hexagonal touching pattern | FCC {111}, HCP {0001}

Densest plane | The plane with highest planar density in that structure | BCC {110}, SC {100}

Close-packed direction | A direction along which atoms touch most closely | FCC $\langle 110 \rangle$, BCC $\langle 111 \rangle$, SC $\langle 100 \rangle$

Coordination number | Number of nearest neighbors around one atom or ion | FCC = 12, BCC = 8, NaCl = 6

This is why BCC can have a closest-packed direction, $\langle 111 \rangle$, without having a true close-packed plane.

Short Memory Version

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FCC:
  CN = 12
  APF = 0.74
  close-packed plane = {111}
  close-packed direction = <110>
  stacking = ABCABC

HCP:
  CN = 12
  APF = 0.74
  close-packed plane = {0001}
  close-packed direction = <11-20>
  stacking = ABAB

BCC:
  CN = 8
  APF = 0.68
  no true close-packed plane
  densest plane = {110}
  closest-packed direction = <111>

Simple Cubic:
  CN = 6
  APF = 0.52
  no true close-packed plane
  densest plane = {100}
  nearest-neighbor direction = <100>

Rock Salt:
  CN = 6 for each ion
  4 formula units per unit cell
  two interpenetrating FCC sublattices
  cations fill octahedral sites
  best described as octahedral ionic coordination, not simple metallic close
  packing

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Common Mistakes

1. Saying BCC has close-packed planes.

More accurate: BCC has no true close-packed plane. Its densest planes are {110}, and its closest-packed directions are <111>.

2. Saying FCC and HCP are completely different in packing efficiency.

More accurate: FCC and HCP both have APF = 0.74 and coordination number 12. Their main difference is stacking sequence.

3. Treating rock salt as if it were a one-atom FCC metal.

More accurate: Rock salt is an ionic structure with two ion types. It has octahedral coordination, 6:6 coordination, and 4 formula units per cubic unit cell.

4. Confusing coordination number with number of atoms per unit cell.

Example: FCC has 4 atoms per conventional unit cell, but each atom has 12 nearest neighbors.